

# Literature resolution: DSD2P is the Daugavet property

Automated Math Discovery Run `fa_banach_001`

Model: GPT5.6

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**Status: literature already answered.** The source question has a direct affirmative solution in Kadets's Theorem 2.3 (2020 preprint; 2021 publication).

## Source question

Haller, Pirk, and Põldvere prove stability of the diametral strong diameter two property (DSD2P) under  $\ell_1$ -sums. The final paragraph of their paper asks whether DSD2P and the Daugavet property coincide [1, p. 4, final paragraph].

In their formulation,  $X$  has DSD2P if for every finite convex combination  $C$  of relatively weakly open subsets of  $B_X$ , every  $x \in C$ , and every  $\varepsilon > 0$ , there is  $y \in C$  such that

$$\|x - y\| \geq 1 + \|x\| - \varepsilon.$$

## Exact later answer

Kadets states that the aim of his paper is to prove the missing converse and then gives the following result.

**Answering theorem** (Kadets, Theorem 2.3). *Let  $X$  be a Banach space with the diametral strong diameter two property. Then  $X$  has the Daugavet property.*

This is Theorem 2.3 on p. 3 of the arXiv version [2, Theorem 2.3, p. 3]. Since the implication from the Daugavet property to DSD2P was already known, the theorem proves equivalence and answers the source question affirmatively.

The proof reduces the Daugavet slice characterization to DSD2P. Given  $x \in S_X$  and a slice  $S$ , Kadets forms a symmetric convex combination of two smaller slices containing a norm-neighborhood of zero, applies DSD2P at a small multiple  $rx$ , and uses quasi-codirectedness to obtain  $s \in S$  with  $\|x - s\| > 2 - \varepsilon$ .

## Bibliographic identification

The answering preprint is arXiv:2008.07258. Its final publication is V. Kadets, *The diametral strong diameter 2 property of Banach spaces is the same as the Daugavet property*, Proc. Amer. Math. Soc. 149 (2021), no. 6, 2579–2582, DOI 10.1090/proc/15448.

## References

- [1] R. Haller, K. Pirk, and M. Põldvere, *Diametral strong diameter two property of Banach spaces is stable under direct sums with 1-norm*, Acta Comment. Univ. Tartu. Math. 20 (2016), 101–105; arXiv:1603.08082.
- [2] V. Kadets, *The diametral strong diameter 2 property of Banach spaces is the same as the Daugavet property*, Proc. Amer. Math. Soc. 149 (2021), 2579–2582; arXiv:2008.07258.